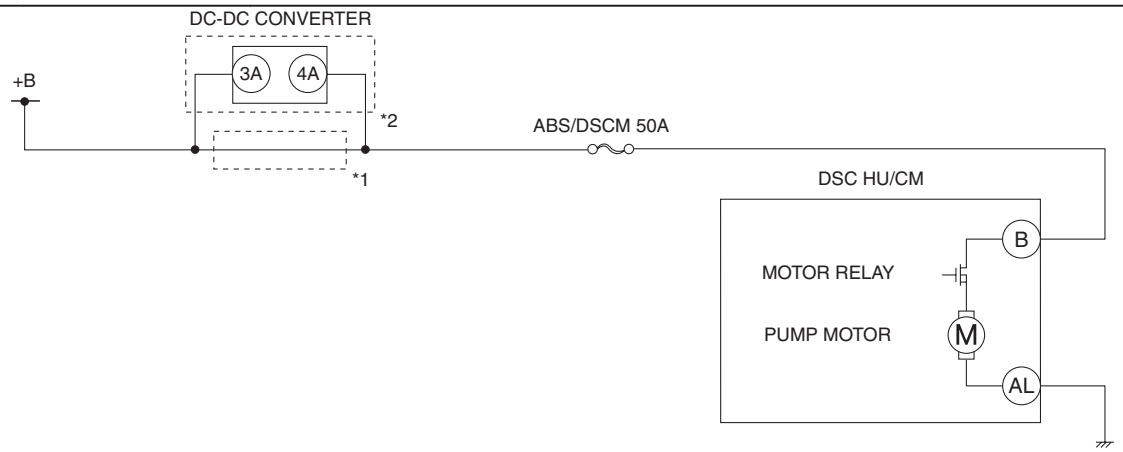
DTC C0020:11/C0020:12/C0020:13/C0020:71 [DSC HU/CM]

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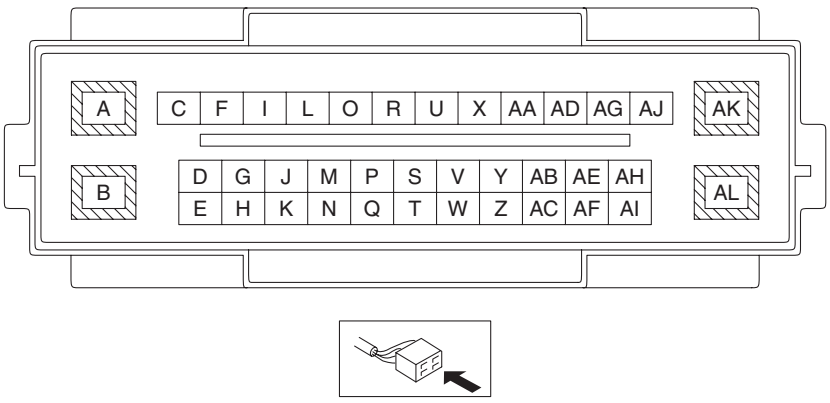
DTC	**C0020:11** **C0020:12** **C0020:13** **C0020:71**	**Pump motor, motor relay**
DETECTION CONDITION		• C0020:11 — Motor relay signal does not correspond to DSC HU/CM OFF signal • C0020:12 — DSC motor monitor signal does not correspond to DSC HU/CM OFF signal — DSC motor monitor signal does not correspond to DSC HU/CM ON signal • C0020:13 — Motor relay signal does not correspond to DSC HU/CM ON signal • C0020:71 — DSC HU/CM motor monitor OFF signal is not input within specified time limit or more when motor signal is switched from ON to OFF by DSC HU/CM
FAIL-SAFE FUNCTION		Refer to DTC TABLE. (See DTC TABLE [DSC HU/CM].)
POSSIBLE CAUSE		**Without i-ELOOP** • Fuse (ABS/DSCM 50A) malfunction • Open or short circuit in wiring harness between battery and DSC HU/CM terminal B • Open circuit in wiring harness between DSC HU/CM terminal AL and body ground • Poor connection at each connector • DSC HU/CM malfunction — Motor relay in DSC HU/CM has sticking, open or short circuit — Pump motor in DSC HU/CM has sticking, open or short circuit **With i-ELOOP** • Fuse (ABS/DSCM 50A) malfunction • Open or short circuit in wiring harness between battery and DC-DC converter terminal 3A • Open circuit in wiring harness between DSC HU/CM terminal AL and body ground • Poor connection at each connector • DSC HU/CM malfunction — Motor relay in DSC HU/CM has sticking, open or short circuit — Pump motor in DSC HU/CM has sticking, open or short circuit



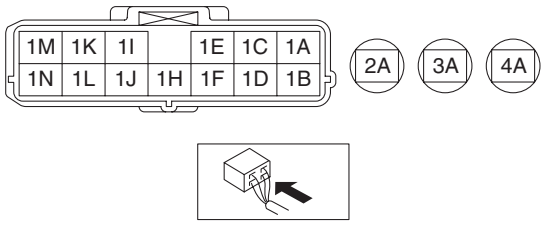
*1: WITHOUT i-ELOOP

*2: WITH i-ELOOP

DSC HU/CM
WIRING HARNESS-SIDE CONNECTOR



DC-DC CONVERTER
WIRING HARNESS-SIDE CONNECTOR



Diagnostic Procedure
Without i-ELOOP

Step	Inspection		Action
1	**INSPECT FUSE CONDITION** • Is the fuse (ABS/DSCM 50A) normal?	Yes	Go to the next step.
		No	• If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between fuse and DSC HU/CM terminal B. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. Go to Step5.
2	**VERIFY PUMP MOTOR OPERATION** • Using the M-MDS, perform the simulation inspection for pump motor. (See SIMULATION INSPECTION [DSC HU/CM].) • Does the pump motor operate?	Yes	Go to Step5.
		No	Go to the next step.
3	**INSPECT MOTOR RELAY POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT** • Switch the ignition OFF. • Disconnect the DSC HU/CM connector. • Switch the ignition ON (engine off). • Measure the voltage between DSC HU/CM terminal B (wiring harness-side) and body ground. • Is the voltage **10 V or more**?	Yes	Go to the next Step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between battery and DSC HU/CM terminal B. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step5.

Step	Inspection		Action
4	**INSPECT PUMP MOTOR GROUND CIRCUIT FOR OPEN CIRCUIT OR POOR GROUND** • Switch the ignition OFF. • Disconnect the DSC HU/CM connector. • Inspect for continuity between DSC HU/CM terminal AL (wiring harness-side) and body ground. • Is the resistance **0—1 ohm**?	Yes	Go to the next step.
		No	• If the open circuit could be detected in the wiring harness: — Refer to the wiring diagram and verify whether or not there is a common connector between DSC HU/CM terminal AL and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. — Go to the next step. • If the resistance is out of specification: — Refer to the wiring diagram and verify whether or not there is a common connector between DSC HU/CM terminal AL and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a poor ground. — Go to the next step.
5	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Reconnect all disconnected connectors. • Using the M-MDS, clear the DTC from the DSC HU/CM. (See CLEARING DTC [DSC HU/CM].) • Start the engine and drive the vehicle at 10 km/h {6.2 mph} or more. • Using the M-MDS, perform the DSC HU/CM DTC inspection. (See DTC INSPECTION [DSC HU/CM].) • Is the same DTC present?	Yes	Repeat the inspection from Step1. If the malfunction recurs, replace the DSC HU/CM, then go to the next step. (See DSC HU/CM REMOVAL/INSTALLATION [L.H.D.].) (See DSC HU/CM REMOVAL/INSTALLATION [R.H.D.].)
		No	Go to the next step.
6	**VERIFY NO DTC IS PRESENT** • Are any DTCs present?	Yes	Go to applicable DTC inspection. (See DTC TABLE [DSC HU/CM].)
		No	DTC troubleshooting completed.

With i-ELOOP

Step	Inspection		Action
1	**INSPECT FUSE CONDITION** • Is the fuse (ABS/DSCM 50A) normal?	Yes	Go to the next step.
		No	• If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between fuse and DSC HU/CM terminal B. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. Go to Step5.
2	**VERIFY PUMP MOTOR OPERATION** • Using the M-MDS, perform the simulation inspection for pump motor. (See SIMULATION INSPECTION [DSC HU/CM].) • Does the pump motor operate?	Yes	Go to Step5.
		No	Go to the next step.
3	**INSPECT MOTOR RELAY POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT** • Switch the ignition OFF. • Disconnect the DSC HU/CM connector. • Switch the ignition ON (engine off). • Measure the voltage between DSC HU/CM terminal B (wiring harness-side) and body ground. • Is the voltage **10 V or more**?	Yes	Go to the next Step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector in the followings: • Battery—DC-DC converter terminal 3A • DC-DC converter terminal 4A—DSC HU/CM terminal B **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step5.

Step	Inspection		Action
4	**INSPECT PUMP MOTOR GROUND CIRCUIT FOR OPEN CIRCUIT OR POOR GROUND** • Switch the ignition OFF. • Disconnect the DSC HU/CM connector. • Inspect for continuity between DSC HU/CM terminal AL (wiring harness-side) and body ground. • Is the resistance **0—1 ohm**?	Yes	Go to the next step.
		No	• If the open circuit could be detected in the wiring harness: — Refer to the wiring diagram and verify whether or not there is a common connector between DSC HU/CM terminal AL and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. — Go to the next step. • If the resistance is out of specification: — Refer to the wiring diagram and verify whether or not there is a common connector between DSC HU/CM terminal AL and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a poor ground. — Go to the next step.
5	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Reconnect all disconnected connectors. • Using the M-MDS, clear the DTC from the DSC HU/CM. (See CLEARING DTC [DSC HU/CM].) • Start the engine and drive the vehicle at 10 km/h {6.2 mph} or more. • Using the M-MDS, perform the DSC HU/CM DTC inspection. (See DTC INSPECTION [DSC HU/CM].) • Is the same DTC present?	Yes	Repeat the inspection from Step1. If the malfunction recurs, replace the DSC HU/CM, then go to the next step. (See DSC HU/CM REMOVAL/INSTALLATION [L.H.D.].) (See DSC HU/CM REMOVAL/INSTALLATION [R.H.D.].)
		No	Go to the next step.
6	**VERIFY NO DTC IS PRESENT** • Are any DTCs present?	Yes	Go to applicable DTC inspection. (See DTC TABLE [DSC HU/CM].)
		No	DTC troubleshooting completed.